Dependence Maps: A Graphical Tool for Visualizing Dependence

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Motivation

- Linear relationships are easy to visualize
  - Scatter plots
  - QQ plots
  - Et cetera
- Often not so for nonlinear, nonmonotone relationships

Example 1: Straight Linear Correlation

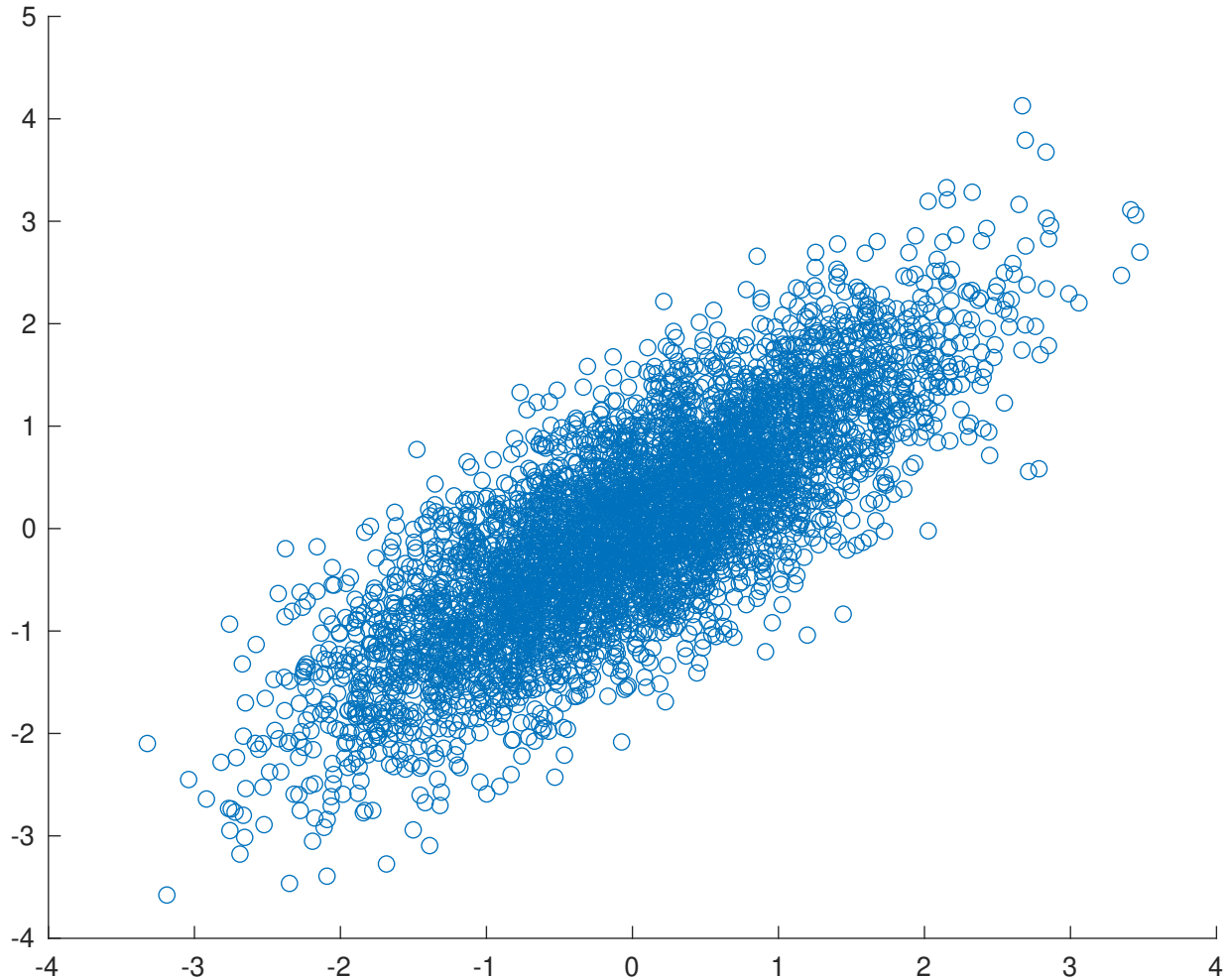


Figure: Bi-Variate Normal, $\rho = .8$, $n = 5,000$

Example 2: Tail Dependence

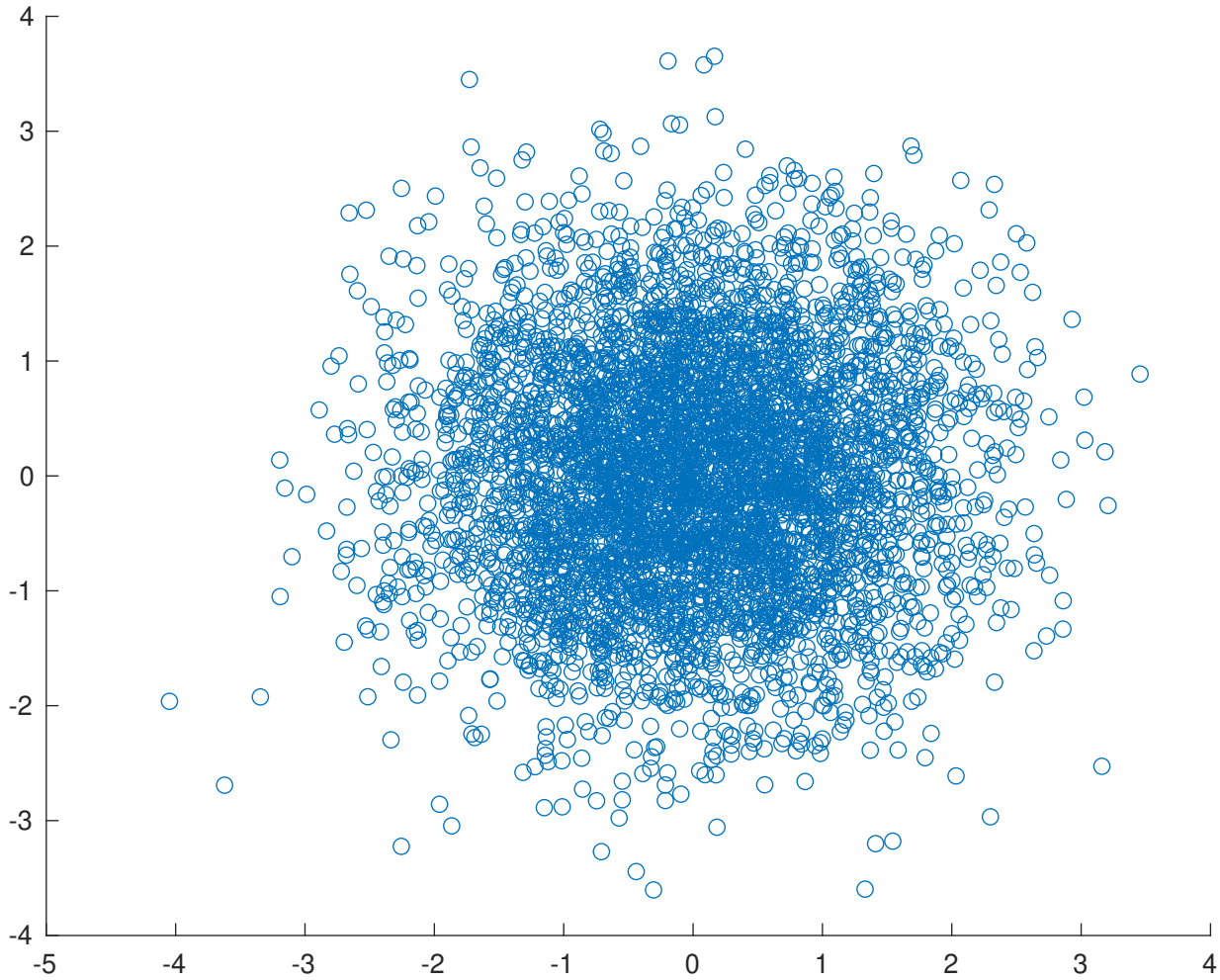


Figure: Bi-Variate t-Distribution, $\rho = 0$, $\tau = 3$, $n = 5,000$

Dependence Maps

Graphical Tool for Analyzing Dependence

- Graphical tool for comprehensive view dependence

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- Graphical tool for comprehensive view dependence
- Computationally straightforward
- Leverages recent advances in dependence measures based on emprical copula process
- Illustrative applications in N.A. power & nat gas markets

Copula-Based Measures of Dependence

Relevant Literature

- Tests of independence among continuous random vectors based on Cramér–von Mises functionals of the empirical copula process, Ivan Kojadinovic, Mark Holmes, *Journal of Multivariate Analysis*, 2009
- Non-parametric weighted tests for independence based on empirical copula process, Ivan Medovikov, *Journal of Statistical Computation and Simulation*, 2014
- A New Measure of Vector Dependence, with Applications to Financial Risk and Contagion, Ivan Medovikov, Artem Prokhorov, *Journal of Financial Econometrics*, 2017

The Copula Function

A Complete Picture of Dependence

- For some continuous X, Y
- With marginals $F(X)$, $G(Y)$ and joint d.f. $H(X, Y)$

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$$H(X, Y) = C(F(X), G(Y)), \tag{1}$$

- Where $F(X)$, $F(X) \sim [0, 1]$, and
- $C(u, v) : [0, 1]^2 \to [0, 1]$ is the *copula* of X, Y

Independence Copula

Mutual Independence of X, Y

- *Independence copula* uniquely characterizes lack of *any* statistical relationship:

$$I(u, v) = uv \tag{2}$$

- Departure from $I(u, v)$ completely characterizes statistical relationship
- *Dependence Maps*: visualize difference between $C(u, v)$ and $I(u, v)$

The Empirical Copula

Nonparametric Estimator

- Given i.i.d. values $(X_1, Y_1), ..., (X_n, Y_n)$

$$\hat{C}(u, v) = \frac{1}{n} \sum_{i=1}^{n} I(\hat{u}_i \leq u, \hat{v}_i \leq v) \tag{3}$$

where

$$\hat{u}_i = \frac{1}{n} \sum_{j=1}^{n} I(X_j \leq X_i), \tag{4}$$

$$\hat{v}_i = \frac{1}{n} \sum_{j=1}^{n} I(Y_j \leq Y_i) \tag{5}$$

Dependence Map

Visualization of Dependence

- Heat map to represent differences:

$$|\hat{C}(u, v) - uv| \tag{6}$$

  on the $[0, 1]^2$ support

- Deviation from uv on the map indicates *dependence at that region* of joint distribution (e.g. median, upper, lower tail, etc)

Example 1: Independence

Independent Random Variables

- Marginally standard normal
- Jointly normal (Gaussian Copula), $\rho = 0$

DepMap: Independence

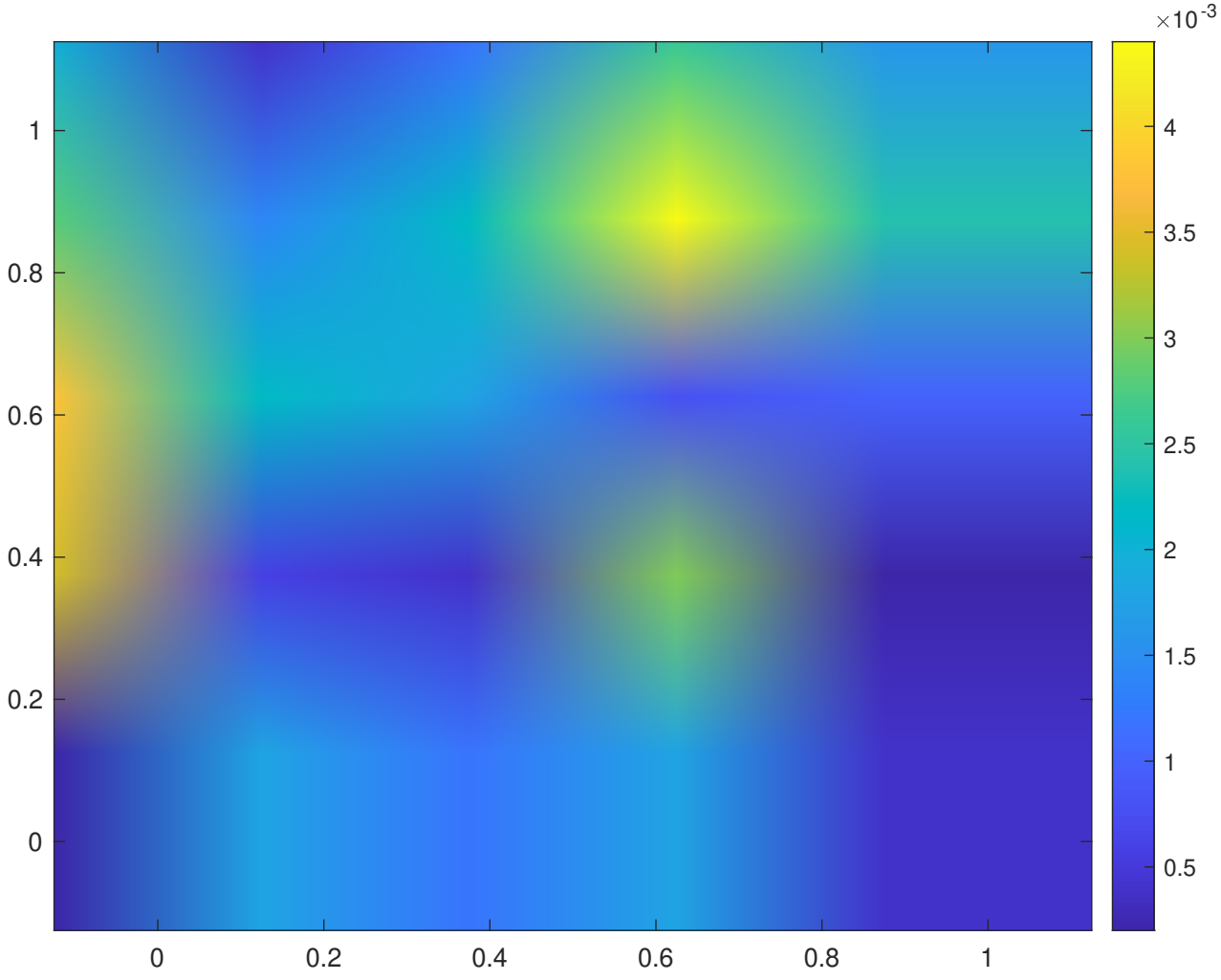
Tabular Representation

	0.1	**0.3**	**0.5**	**0.7**	**0.9**
0.1	0.0002	0.0018	0.0012	0.0018	0.0004
0.3	0.0034	0.0006	0.0004	0.0030	0.0002
0.5	0.0038	0.0022	0.0018	0.0008	0.0010
0.7	0.0028	0.0014	0.0022	0.0044	0.0024
0.9	0.0020	0.0004	0.0012	0.0026	0.0016

Table: Numerical values table.

DepMap: Independence

Graphical Representation



DepMap: Linear Correlation

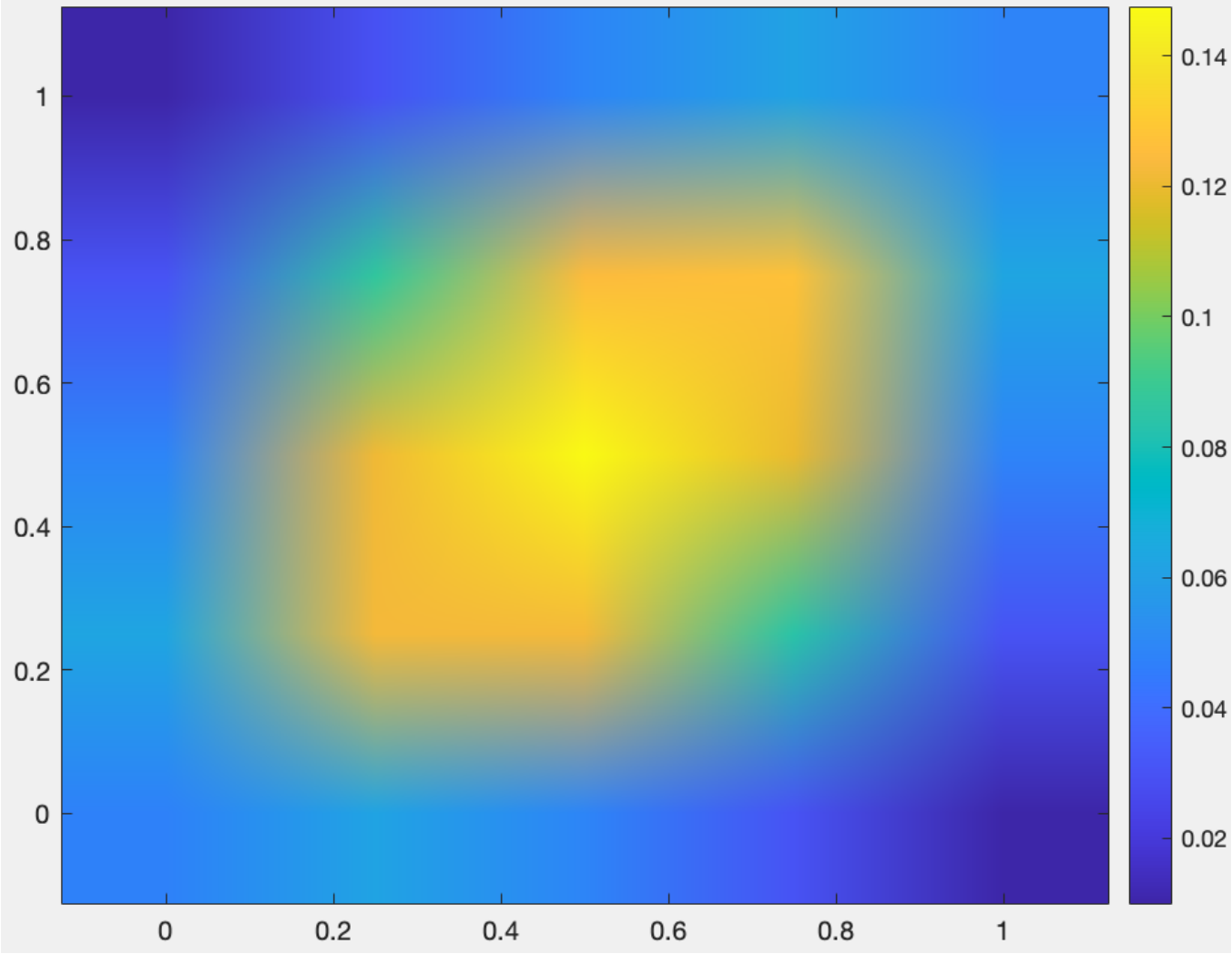
Example 1: Bi-Variate Gaussian, $\rho = .8$

	0.1	**0.3**	**0.5**	**0.7**	**0.9**
0.1	0.0476	0.0616	0.0488	0.0300	0.0100
0.3	0.0632	0.1224	0.1224	0.0838	0.0300
0.5	0.0490	0.1216	0.1476	0.1200	0.0482
0.7	0.0300	0.0866	0.1246	0.1268	0.0628
0.9	0.0100	0.0298	0.0488	0.0616	0.0486

Table: Generated table with values from the provided image.

DepMap: Linear Dependence

Example 1: Graphical Representation



DepMap: Tail Dependence

Example 2: Bi-Variate t, $\rho = 0$, $\tau = 3$

	0.1	**0.3**	**0.5**	**0.7**	**0.9**
0.1	5.6569	2.366	0.4714	4.5262	6.7568
0.3	1.9545	0.87547	0.18516	2.357	2.9832
0.5	1.3199	0.67893	0.45255	1.0493	0.37712
0.7	4.9377	1.4816	1.7131e-14	2.2223	2.8803
0.9	7.0711	2.0574	0.94281	4.5262	7.3853

Table: Generated table with values from the second image.

DepMap: Tail Dependence

Example 2: Graphical Representation

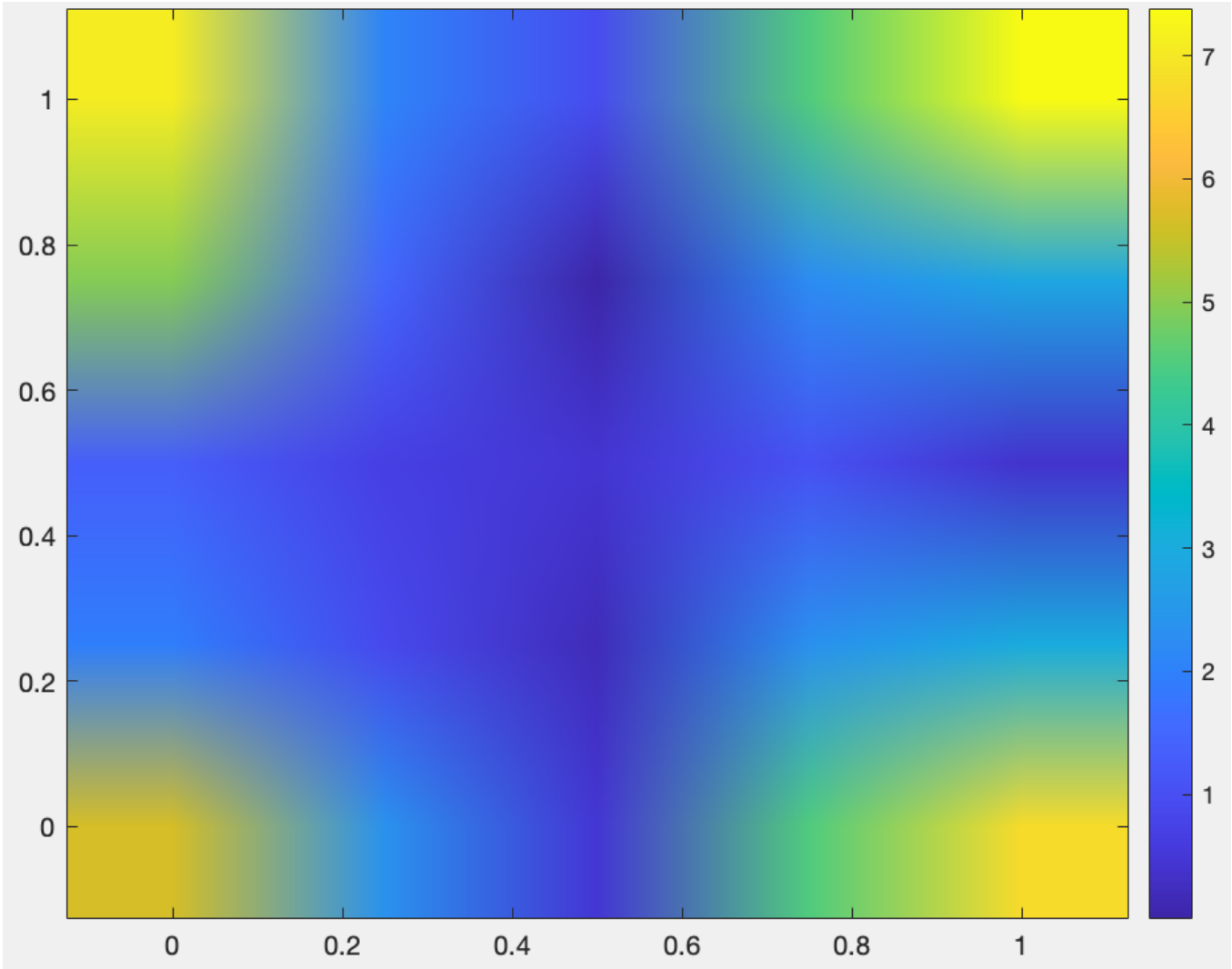


Figure: Caption

Scatter: Tail Dependence

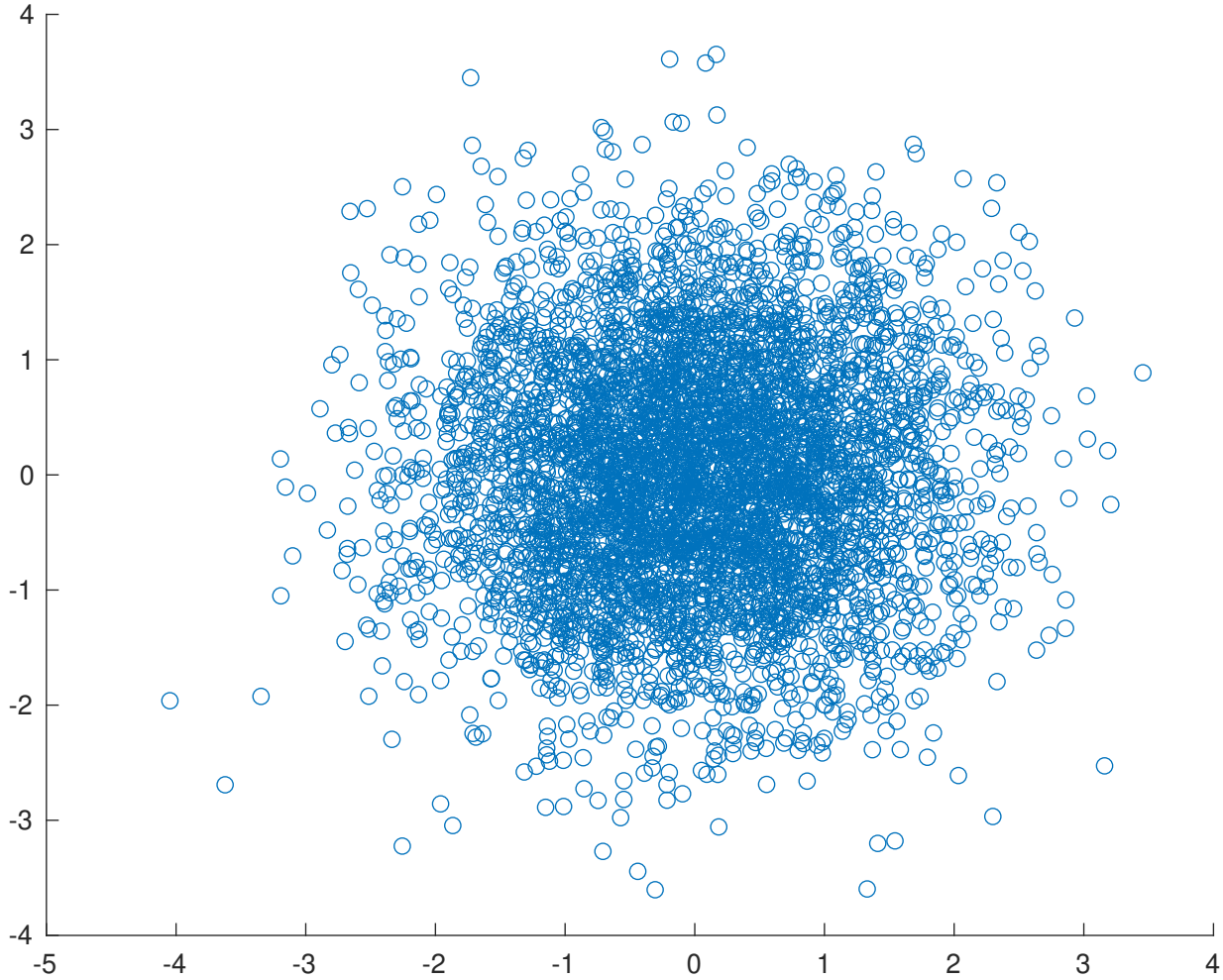


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Application: Wholesale Power & Natural Gas

New York Independent System Operator

- Day-ahead and real-time (balancing) market for wholesale power

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New York Independent System Operator

- Day-ahead and real-time (balancing) market for wholesale power
- Operates at up to 5 minute intervals, hundreds of locations (LMPs)
- Price drivers include loads, weather, gas prices
- Use DepMaps to understand price signals

Gas Price vs DA Electricity Price

N.Y.C.

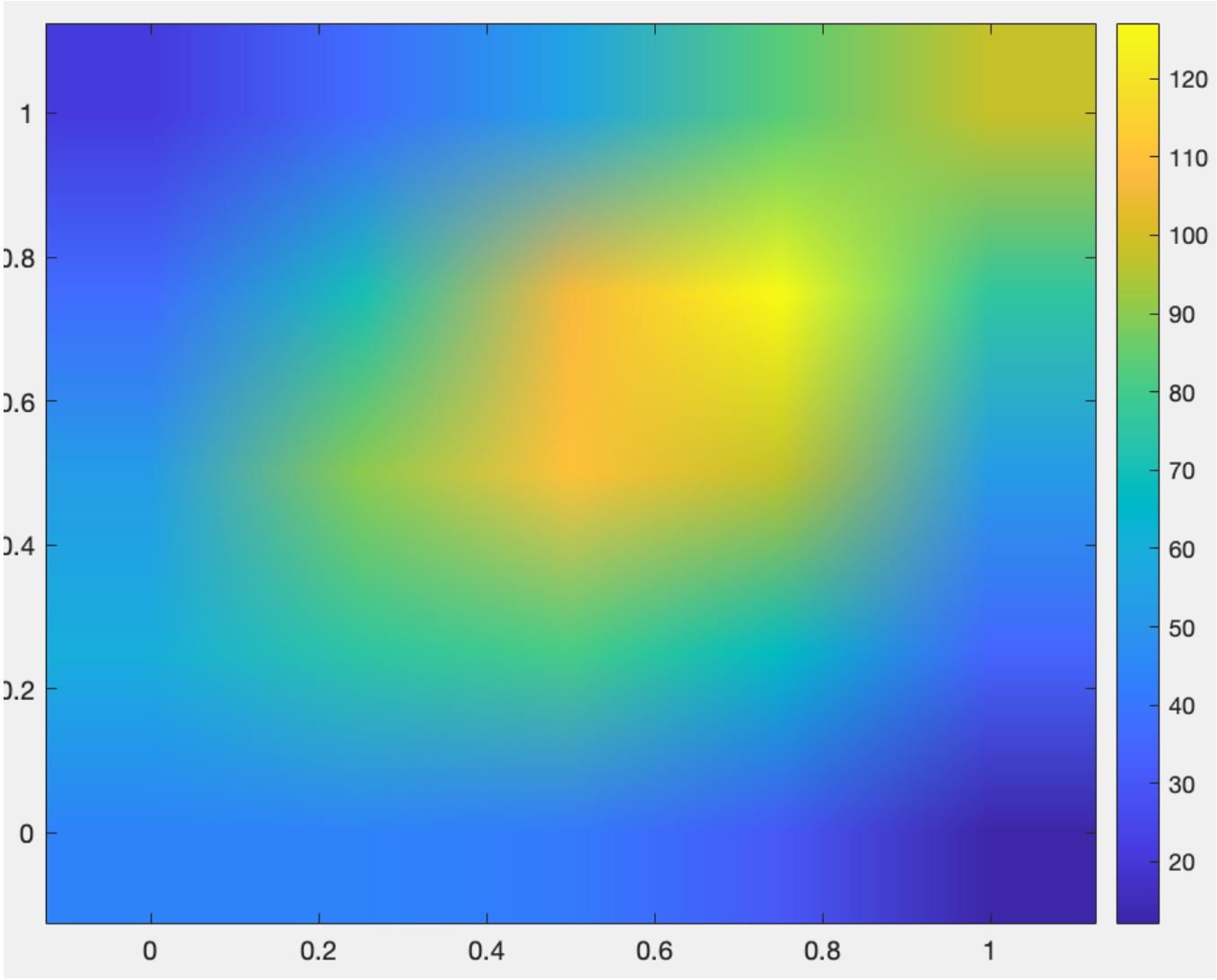


Figure: DepMap: TETCO M3 vs N.Y.C. DALMP

Gas Price vs DA Electricity Price

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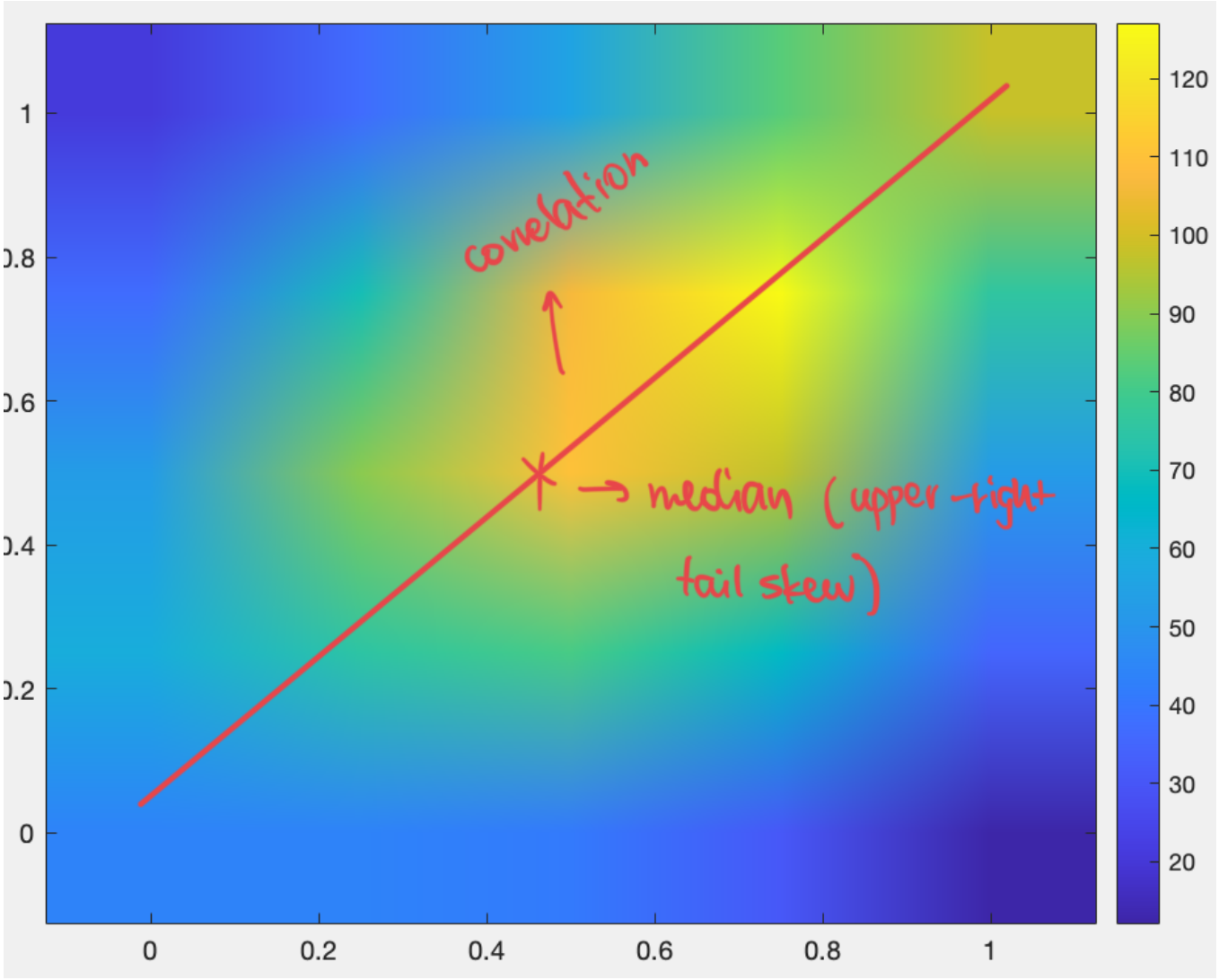


Figure: DepMap: TETCO M3 vs N.Y.C. DALMP

Gas Price vs DA Electricity Price

Insights

- Positive linear association between TETCO M3 and energy prices
- High gas price is a strong signal of high power price
- Low gas price is not a good predictor of low power price

Temperature vs DALMP

N.Y.C.

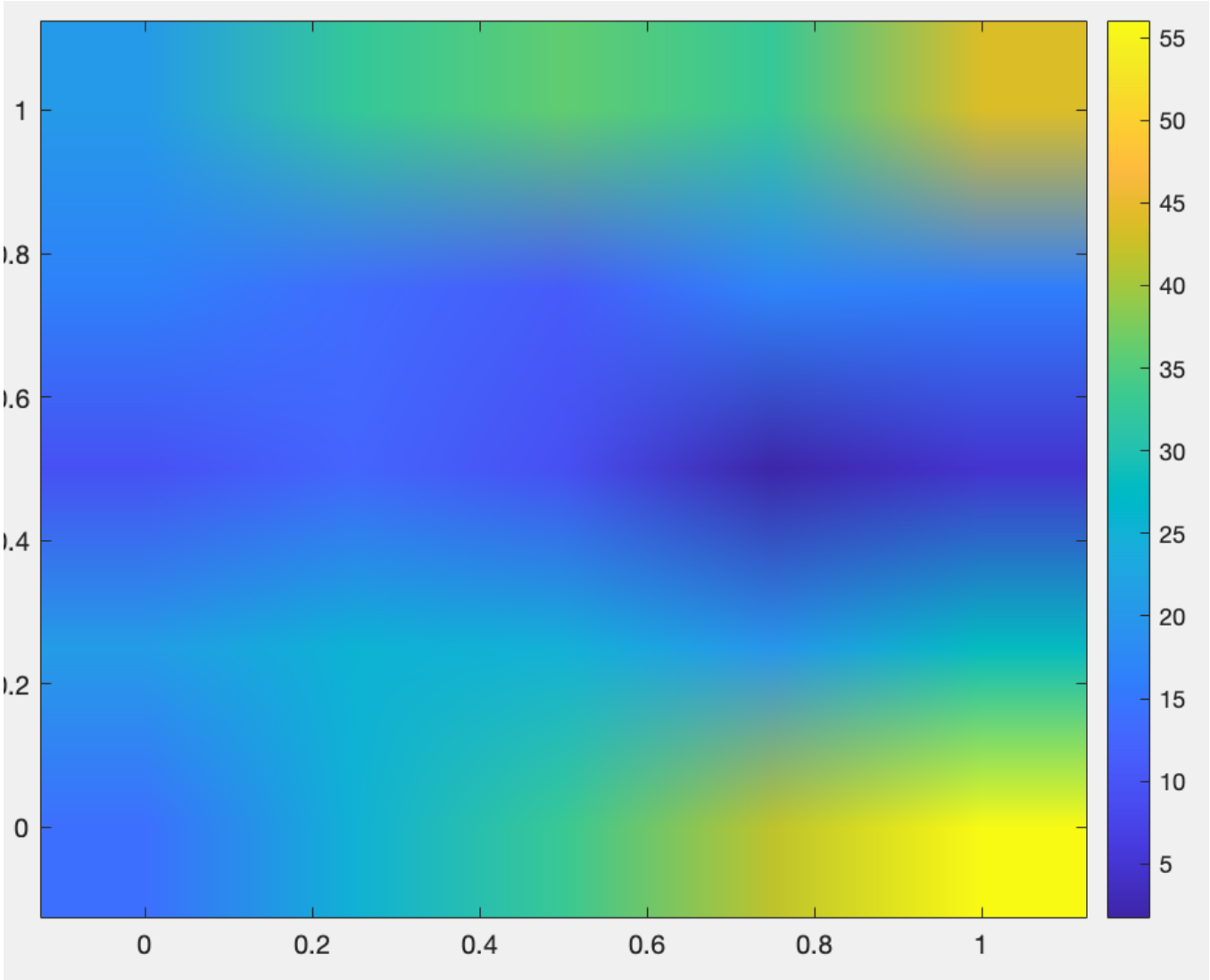


Figure: Caption

Temperature vs DALMP

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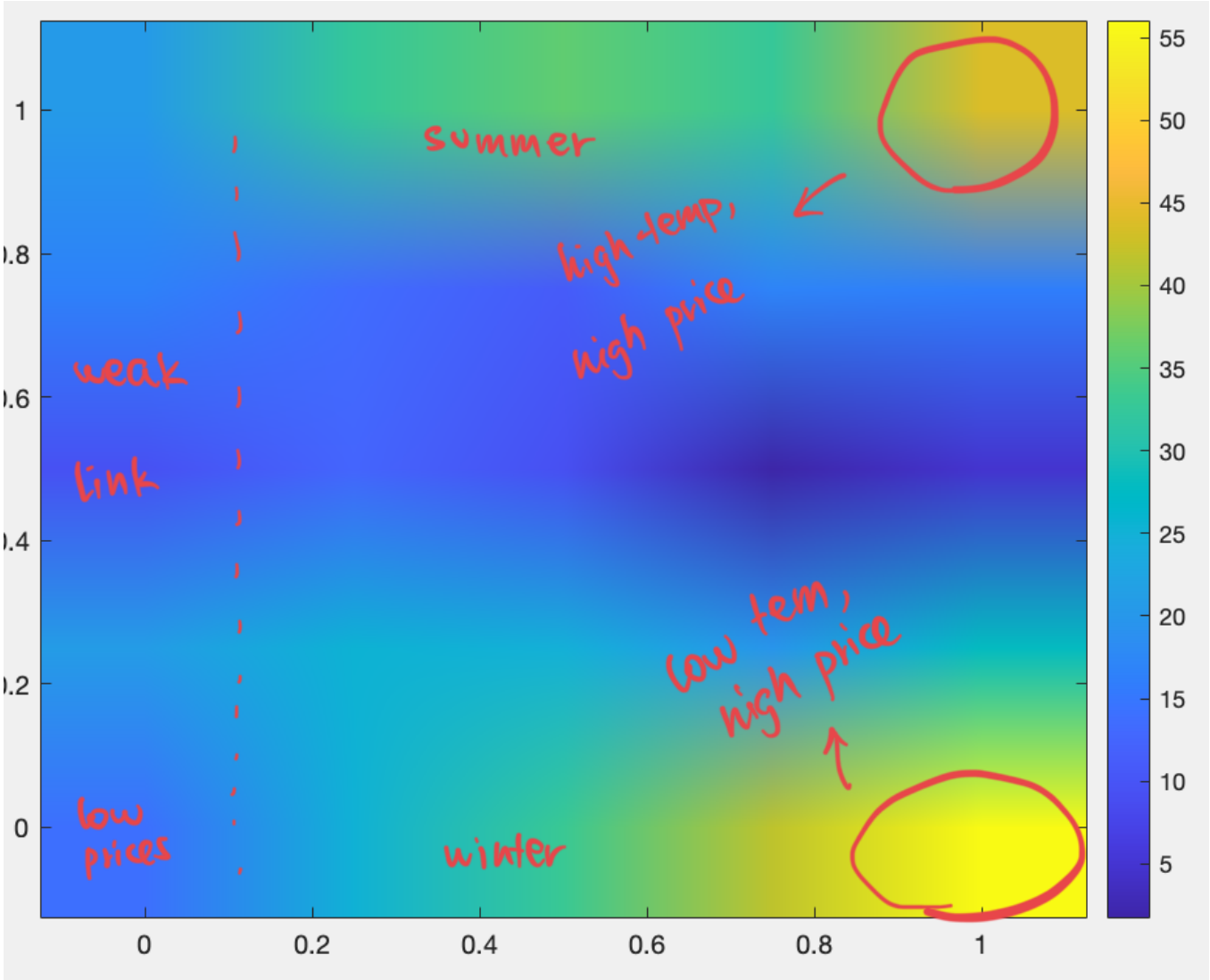


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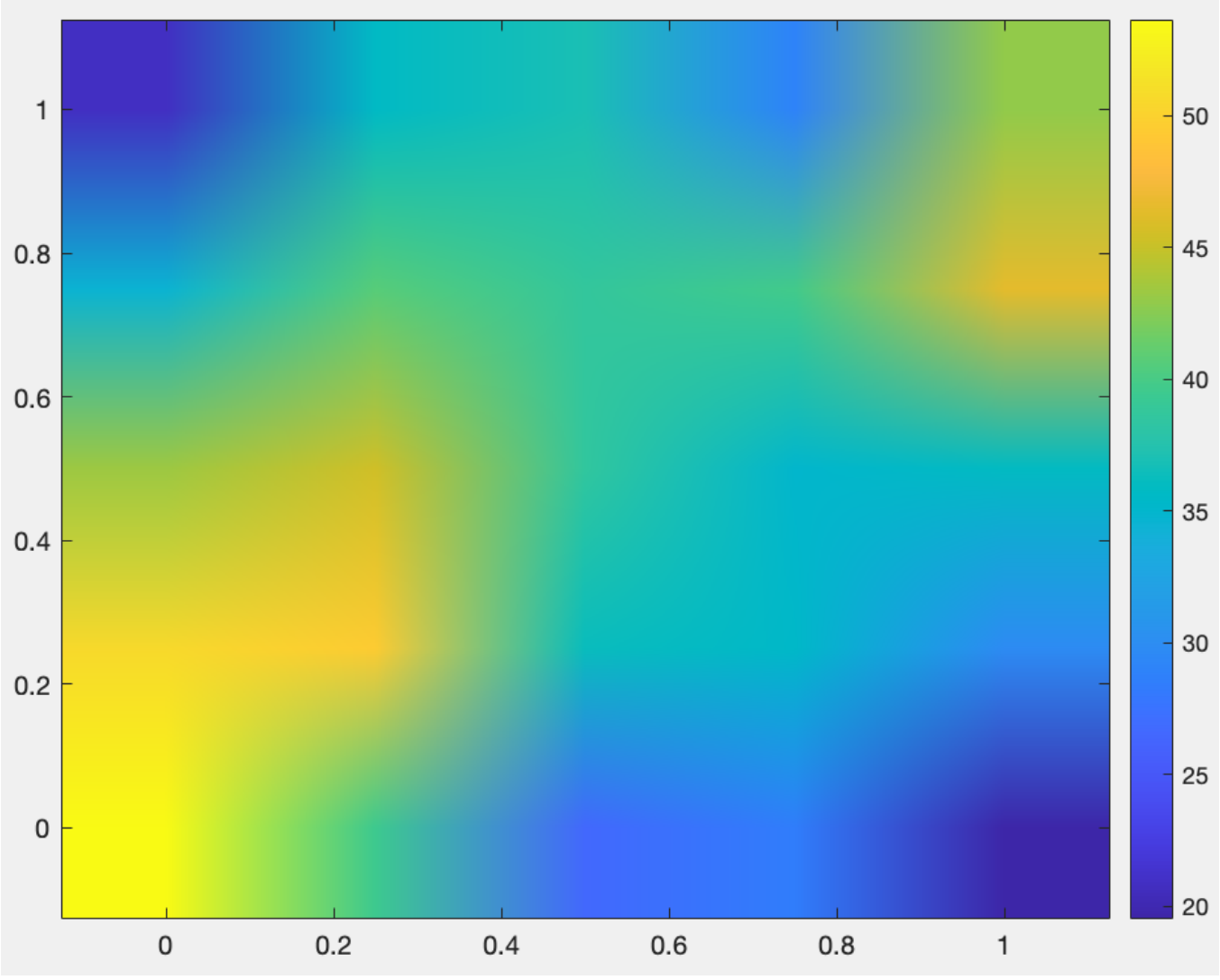
Temperature vs DALMP

Insights

- Seasonality clear in the data
- Temperature extremes translate into high but not low price extremes
- Effect of extremely-low temperature outweighs that of extremely-high temperature
- Weak links between low prices and temperature

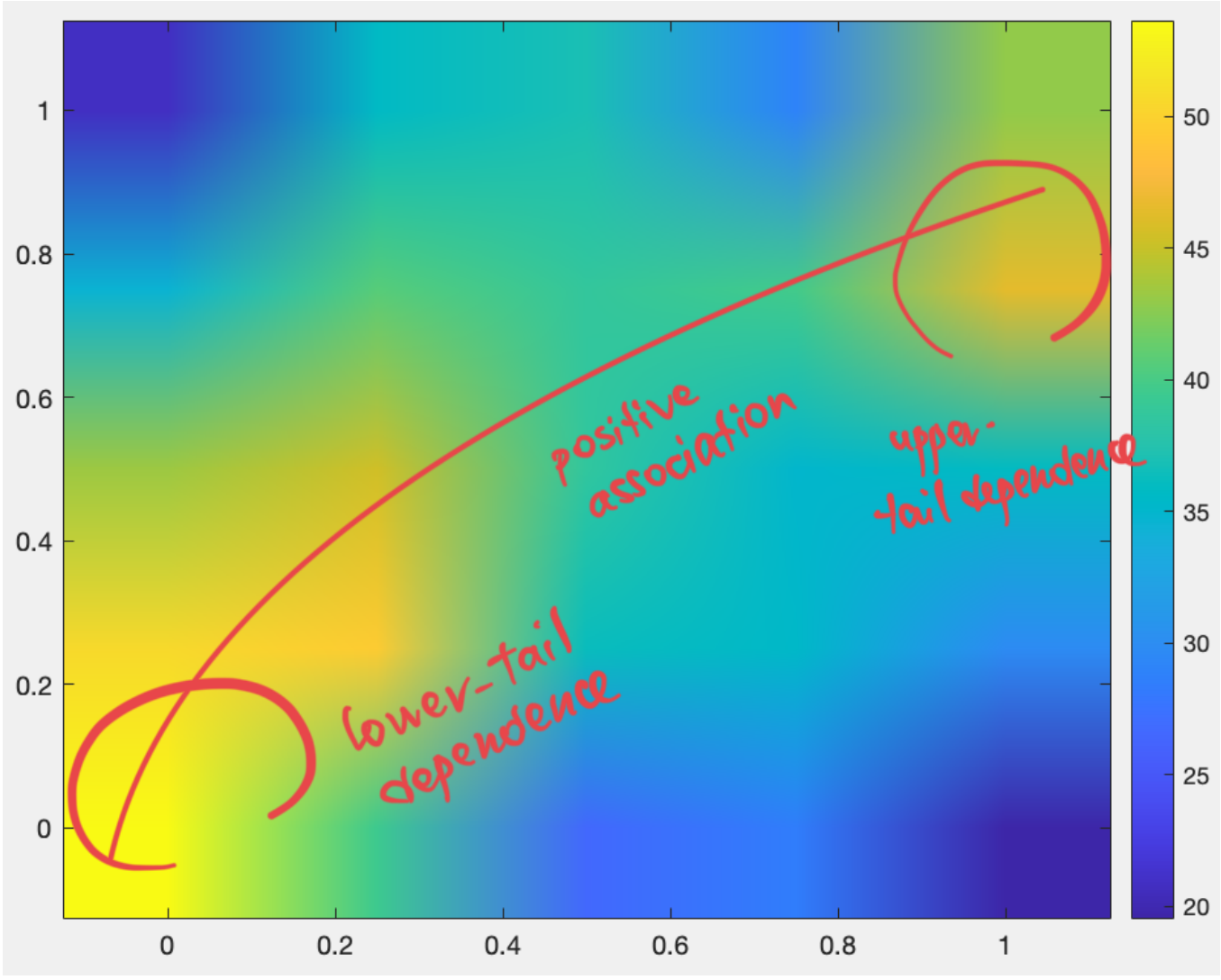
Load vs DALMP

N.Y.C.



Load vs DALMP

N.Y.C.



Load vs CALMP

N.Y.C.

- Positive, non-linear relationship between load and LMP
- Asymmetric tail dependence
  - Extremely low loads strongly predictive of extremely low prices
  - Extremely high loads less predictive of extremely high prices
  - Weaker relationship around the median

The Vector Version

Multi-Variate Extension

- Can we map dependence between groups of variables?

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Measure of Vector Dependence

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- Let $\mathbf{x} = (X_1, X_2)$ and $\mathbf{y} = (Y_1, Y_2)$
- Vector $\mathbf{x}$ is independent from $\mathbf{y}$ when

$$H(\mathbf{x}, \mathbf{y}) = F(\mathbf{x})G(\mathbf{y}), \tag{7}$$

where $F(\mathbf{x}) = C_x(F_1(X_1), F_2(X_2))$ and $G(\mathbf{y}) = C_y(G_1(Y_1), G_2(Y_2))$

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  where $F(\mathbf{x}) = C_x(F_1(X_1), F_2(X_2))$ and $G(\mathbf{y}) = C_y(G_1(Y_1), G_2(Y_2))$
- Can similarly visualize difference

$$|C - C_x C_y| \tag{8}$$

Alternative Estimators

Copula Sieve MLE

- Efficient estimation of parameters in marginals in semiparametric multivariate models, I. Medovikov, V. Panchenko and A. Prokhorov, *Journal of Computational and Graphical Statistics*, forthcoming